

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4

Total Marks:40

Annexure A

ARTICLE REVIEW

Code of Conduct:

*I **NEELAM NANDINI [192472277]** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: N. Nandini.

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications*.

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

[Selected Article](#)

Article: *Diabetes type 2 classification using machine learning algorithms with up-sampling technique*

Author: Mariwan Ahmed Hama Saeed

Journal: *Journal of Electrical Systems and Information Technology*, Volume 10, Article 8

Publisher: Springer Nature

Year: 2023

DOI: 10.1186/s43067-023-00074-5

Domain: Healthcare / Medical Diagnosis

ML Problem: Classification and prediction of Type 2 diabetes using machine learning algorithms.

CO 4 Topic: Decision Trees / Inductive Learning

Article link: [Springer – Full Article and PDF](#)

The article is open access and the Springer page provides a Download PDF option. The publication date is 8 February 2023.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

[\(a\) Citation, Domain and ML Problem](#)

[Full Citation](#)

Hama Saeed, M. A. (2023). *Diabetes type 2 classification using machine learning algorithms with up-sampling technique*. *Journal of Electrical Systems and Information Technology*, 10, 8. Springer Nature. <https://doi.org/10.1186/s43067-023-00074-5>.

Domain/Application Area

The research belongs to the healthcare and medical diagnosis domain. The application focuses on automatically identifying whether a person is likely to have Type 2 diabetes.

Specific Machine Learning Problem

The problem is a binary classification problem:

- Class 0: Non-diabetic
- Class 1: Diabetic

The study compares four machine learning classifiers:

1. Decision Tree Classifier
2. AdaBoost Classifier
3. Gradient Boosting Classifier
4. Extra Trees Classifier

The objective is to determine which classifier provides better diabetes classification performance using the PIMA and BRFSS datasets.

(b) Key Objective, Research Gap and Proposed Solution

The main objective of the article is to develop and compare machine learning models for Type 2 diabetes classification while addressing the problem of imbalanced medical datasets. Diabetes prediction is important because early identification can help healthcare professionals take preventive measures and reduce complications. The authors identified that previous studies had used many machine learning and deep learning algorithms, but the datasets often contained imbalanced classes, which could negatively affect classification performance. The study therefore focuses on four relatively efficient classification algorithms: Decision Tree, AdaBoost, Gradient Boosting and Extra Trees. Two datasets, PIMA Indian Diabetes and BRFSS, were used to make the evaluation more comprehensive. Before classification, outliers were removed and an up-sampling technique was applied to balance the classes. The datasets were then divided into 80% training and 20% testing data. The models were evaluated using accuracy, precision, recall, F1-score and ROC/AUC. The Extra Trees classifier achieved the strongest overall performance, with ROC/AUC values of approximately 0.96 for PIMA and 0.99 for BRFSS. The study demonstrates that class balancing combined with ensemble tree methods can improve diabetes classification performance.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

(a) ML Techniques Used

The researchers used four supervised machine learning classification algorithms.

1. Decision Tree Classifier

A Decision Tree is an inductive learning algorithm that learns decision rules from training examples.

It creates a tree structure consisting of:

- Root node
- Internal decision nodes
- Branches
- Leaf/output nodes

The algorithm repeatedly divides the dataset according to feature values until a final classification is obtained.

For example:

Glucose > threshold → BMI > threshold → Age > threshold → Diabetic

The study reports ROC values of approximately 0.78 for PIMA and 0.92 for BRFSS for the Decision Tree.

2. AdaBoost Classifier

AdaBoost combines several weak learners to create a stronger classifier. Later learners concentrate more on incorrectly classified observations from previous learners.

The reported ROC values were approximately:

- PIMA: 0.83
- BRFSS: 0.82

3. Gradient Boosting Classifier

Gradient Boosting constructs models sequentially, with each new model attempting to reduce the errors of previous models.

Reported ROC values:

- PIMA: 0.90
- BRFSS: 0.82

4. Extra Trees Classifier

Extra Trees is an ensemble of randomized decision trees. Multiple trees are constructed and their predictions are combined.

The study selected Extra Trees as the best-performing model, reporting ROC/AUC values of approximately:

- PIMA: 0.96
- BRFSS: 0.99

Dataset Description

PIMA Indian Diabetes Dataset

The PIMA dataset was obtained from the National Institute of Diabetes and Digestive and Kidney Diseases.

The dataset contains medical attributes including:

Feature	Description
Pregnancies	Number of pregnancies
Glucose	Plasma glucose concentration
Blood Pressure	Blood pressure
Skin Thickness	Skin-fold thickness
Insulin	Insulin level
BMI	Body Mass Index
Diabetes Pedigree Function	Family-history-related diabetes measure
Age	Patient age
Outcome	Diabetes classification

The article reports approximately 769 records for the dataset used in its description.

BRFSS Dataset

The Behavioral Risk Factor Surveillance System dataset was obtained from the CDC.

It contains 253,680 records and 21 feature variables. Features include factors such as:

- Smoking
- Stroke
- Heart disease
- Physical activity
- Fruit consumption
- Sex
- Education
- Income
- Diabetes status

Data Preparation

The study followed these main preprocessing steps:

Raw Dataset → Outlier Removal → Class Balancing → 80% Training / 20% Testing → ML Classification → Performance Evaluation

The authors observed class imbalance in both datasets. They therefore applied an up-sampling technique to balance the classes. After preprocessing, 80% of the data was randomly used for training and 20% for testing. Python, Jupyter Notebook, Pandas, NumPy, Matplotlib and Scikit-learn were used in the implementation.

(b) Mapping to CO 4 Topics

Topic	Connection with Article
Inductive Learning	Models learn general classification patterns from labelled patient examples.
Decision Trees	Decision Tree Classifier is directly implemented.
Statistical Learning	Accuracy, precision, recall, F1-score and ROC/AUC are used for evaluation.
Ensemble Learning	AdaBoost, Gradient Boosting and Extra Trees combine multiple learners.
Classification	The problem is binary classification of diabetic/non-diabetic patients.

Methodological Strength

A major strength is the use of two different diabetes datasets rather than relying on only one dataset. This gives a broader evaluation of the classification models. The authors also explicitly address class imbalance through up-sampling and evaluate the models using several metrics rather than accuracy alone.

Methodological Limitation

A limitation is the use of a single random 80:20 train-test split rather than repeated cross-validation or external validation. This can make the reported performance dependent on the particular random split. Also, up-sampling before the train-test split can potentially introduce duplicated observations into both training and testing data if not carefully separated, which could lead to optimistic results.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

(b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?

(c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

(a) Key Results

The paper compared four machine learning models on both datasets.

ROC/AUC Comparison

Algorithm	PIMA ROC/AUC	BRFSS ROC/AUC
Decision Tree	0.7888	0.92
AdaBoost	0.83	0.8247
Gradient Boosting	0.90	0.82
Extra Trees	0.96	0.99

The Extra Trees classifier achieved the highest ROC/AUC values for both datasets.

The authors also report overall accuracy values of approximately:

Dataset Reported Best Accuracy

PIMA 89%

BRFSS 96%

For the PIMA dataset, the Extra Trees classifier achieved recognition rates of approximately 92% for the positive class and 87% for the negative class. For BRFSS, Extra Trees and Decision Tree achieved very high recognition rates for the positive class, while the overall results favored Extra Trees.

(b) Critical Evaluation

Are the Results Realistic?

The reported results are technically possible, but they should not automatically be interpreted as clinical-level performance.

The BRFSS dataset is very large, which can help machine learning algorithms identify patterns effectively. However, high performance on a benchmark dataset does not guarantee equivalent performance on hospital patients from different populations.

The PIMA dataset is comparatively small and contains a specific population, so its results may not generalize to all demographic groups.

Potential Biases

Several potential sources of bias exist:

1. Dataset bias

The PIMA dataset represents a specific population rather than the entire global population.

2. Class imbalance

The original datasets were imbalanced. Although up-sampling attempts to correct this, it can affect the distribution of the training data.

3. Sampling bias

The study uses existing datasets rather than prospectively collected clinical data.

4. Evaluation bias

A single 80:20 random split can produce results that vary depending on the particular split.

5. Data leakage risk

When up-sampling is performed, it is important that the test set remains completely independent from duplicated training observations. A stronger protocol would apply balancing only to the training folds.

Additional Experiments That Would Strengthen the Study

The research could be strengthened by:

1. Using 5-fold or 10-fold stratified cross-validation.
2. Performing external validation using an independent hospital dataset.
3. Reporting confidence intervals for accuracy and AUC.
4. Comparing sensitivity and specificity across different decision thresholds.
5. Using SHAP or LIME for model explainability.
6. Testing the model on patients from different countries and demographic groups.
7. Comparing up-sampling with SMOTE, ADASYN and class-weighted learning.

(c) Real-World Applicability

The proposed approach can potentially be used in:

- Hospital decision-support systems
- Diabetes screening applications
- Primary healthcare centres
- Telemedicine platforms
- Patient monitoring systems

- Health-risk assessment applications
- Community-level diabetes screening

A healthcare application could receive patient information such as glucose, BMI, age and blood pressure and produce a preliminary diabetes-risk classification.

However, the model should be treated as a decision-support tool rather than a replacement for a medical professional.

Industry Scaling Challenges

Data availability: Healthcare organisations need large, high-quality and representative patient datasets.

Privacy: Medical information is sensitive and must be protected.

Generalization: A model trained on one population may perform poorly on another.

Integration: ML systems must integrate with Electronic Health Record systems.

Explainability: Doctors need understandable reasons for predictions.

Regulatory requirements: Clinical AI systems require appropriate validation and regulatory approval.

Ethical concerns: Incorrect predictions could result in unnecessary anxiety or missed diagnosis.

Computational requirements: Large healthcare datasets require suitable storage and processing infrastructure.

Task 4: Reflection & Learning Outcome

(a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.

(b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

(a) Three Key Insights Gained

Insight 1 – Decision Trees Learn Rules from Examples

The article demonstrates how a Decision Tree can learn classification rules from labelled patient data.

CO 4 Connection: This directly relates to Inductive Learning, where a general hypothesis is learned from specific training examples.

For example:

High glucose + high BMI + older age → higher diabetes risk

This helped me understand how decision trees transform training data into understandable decision rules.

Insight 2 – Data Preprocessing Can Strongly Affect ML Performance

The research shows that class imbalance can influence classification performance. The authors therefore used up-sampling before training the models.

CO 4 Connection: This connects machine learning algorithms with data preprocessing and training-set preparation.

I learned that selecting an appropriate algorithm alone is not enough. The quality and distribution of the training data can significantly influence the final model.

Insight 3 – Accuracy Alone Is Not Enough

The article evaluates models using accuracy, precision, recall, F1-score and ROC/AUC.

CO 4 Connection: This relates to Statistical Learning and Model Evaluation.

For medical classification, recall and specificity are especially important because false negatives may represent patients who actually have diabetes but are incorrectly classified as healthy.

Therefore, a model with high accuracy is not necessarily the best model for real-world healthcare.

(b) Proposed Novel Extension

Proposed Experiment: Cross-Dataset Explainable Diabetes Prediction

I would extend the research by developing an explainable ensemble diabetes prediction system that combines the PIMA and BRFSS datasets with a newer clinical dataset.

The experiment would compare:

- Decision Tree
- Random Forest
- Extra Trees
- XGBoost
- SVM
- Logistic Regression

The models would be evaluated using stratified 10-fold cross-validation and then tested on an independent external dataset. Instead of using only up-sampling, the experiment would compare SMOTE, class weighting and balanced ensemble methods.

SHAP would then be used to explain the most influential features for each prediction.

Justification

This extension is feasible because the original research already uses publicly available diabetes datasets and tree-based algorithms. The additional methods can be implemented using standard Python machine-learning libraries.

Expected Impact

The experiment could improve:

- Generalization
- Reliability
- Interpretability
- Fairness across populations
- Detection of high-risk patients

Most importantly, it would move the research from simply achieving high benchmark accuracy toward developing a more reliable and explainable clinical decision-support system.

Overall Review and Conclusion

The article provides a useful application of Decision Trees and ensemble learning to Type 2 diabetes classification. Its major contribution is the comparison of Decision Tree, AdaBoost, Gradient Boosting and Extra Trees using both PIMA and BRFSS datasets after addressing class imbalance through up-sampling. The Extra Trees classifier produced the strongest reported performance, with ROC/AUC values of approximately 0.96 for PIMA and 0.99 for BRFSS.

From a CO 4 perspective, the article demonstrates how inductive learning, decision-tree classification, ensemble learning, and statistical evaluation measures can be applied to a real-world healthcare problem. However, the reported performance should be interpreted cautiously because benchmark results do not necessarily represent clinical performance. Cross-validation, external validation, confidence intervals and explainability techniques would make the research more robust.

Overall, the article shows that machine learning can support early diabetes risk classification, but real-world deployment requires representative data, rigorous validation, privacy protection, interpretability and clinical oversight.

References

1. Hama Saeed, M. A. (2023). *Diabetes type 2 classification using machine learning algorithms with up-sampling technique*. Journal of Electrical Systems and Information Technology, 10, 8. DOI: 10.1186/s43067-023-00074-5.
2. National Institute of Diabetes and Digestive and Kidney Diseases. Pima Indians Diabetes Dataset. The article identifies the PIMA dataset as originating from NIDDK.
3. Centers for Disease Control and Prevention. Behavioral Risk Factor Surveillance System (BRFSS). The article identifies BRFSS as the source of its second dataset.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand